

HDOC 14

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Batch: 16

1. Algorithm

1. Start
2. Declare variables for choice, num, temp, rem, isUnique, and an array freq[10] initialized to 0.
3. Display the menu options to the user.
4. Read the user's choice.
5. If choice is 1 or 2, prompt the user to input a positive integer num. Validate that $\text{num} > 0$.
6. Use a switch statement based on the value of choice.
7. Case 1 (Check unique digits):
 - Initialize a tracking array `seen[10] = {0}`.
 - Set `isUnique = 1` and `temp = num`.
 - Loop while (`temp > 0`):
 - Calculate `rem = temp % 10`.
 - If `seen[rem] == 1`, set `isUnique = 0` and break the loop.
 - Else, set `seen[rem] = 1`.
 - Update `temp = temp / 10`.
 - If `isUnique == 1`, display "Unique", else display "Not Unique".
 - Break.
8. Case 2 (Digit frequencies):
 - Initialize all elements of `freq[10]` to 0.

- Set temp = num.
- Loop while (temp > 0):
 - Calculate rem = temp % 10.
 - Increment freq[rem]by 1.
 - Update temp = temp / 10.
- Loop ifrom 0 to 9: display freq_i = freq[i].
- Break.

9. Default Case: Display an "Invalid choice" error message.

10. Stop

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num)	Expected Output
TC-01	Valid inputs (Unique digits)	Choice: 1 Num: 12345	Output: Unique
TC-02	Valid inputs (Repeating digits)	Choice: 1 Num: 12234	Output: Not Unique
TC-03	Frequency tracking (Option 2)	Choice: 2 Num: 112333	freq_1=2, freq_2=1, freq_3=3, rest=0
TC-04	Single digit number (Option 1)	Choice: 1 Num: 7	Output: Unique
TC-05	Invalid negative number input	Choice: 1 Num: -90	Error: Please enter a strictly positive integer.
TC-06	Invalid menu choice	Choice: 3	Invalid choice. Please choose 1 or 2.

3. C Program Source Code

```
#include <stdio.h>

int main() {
    int choice;
    long long num, temp;
    int rem, i;
```

```

int isUnique;
int seen[10] = {0};
int freq[10] = {0};

// Display Menu
printf("--- Digit Operations Menu ---\n"); printf("1. Check whether all digits
are unique\n");
printf("2. Find the frequency of each digit (0-9)\n"); printf("Enter your choice (1 or 2):
");
scanf("%d", &choice);

// Validate Choice and Take Number Input if (choice == 1 ||
choice == 2) {
    printf("Enter a positive integer: "); scanf("%lld", &num);

    if (num <= 0) {
        printf("Error: Please enter a strictly positive integer.\n"); return 1; // Exit program due to invalid
        input
    }
}

// Switch Case Logic switch (choice) {
    case 1:
        // Check for unique digits temp = num;
        isUnique = 1;

        // Re-initialize seen array to 0 just in case for(i = 0; i < 10; i++) seen[i] =
        0;

        while (temp > 0) { rem = temp
            % 10;
            if (seen[rem] == 1) {
                isUnique = 0; // Digit repeated break;
            }
            seen[rem] = 1; // Mark digit as seen temp /= 10;
        }

        if (isUnique) {
            printf("Number: %lld, Output: Unique\n", num);
        } else {
            printf("Number: %lld, Output: Not Unique\n", num);
        }
}

```

```
break;
```

```
case 2:
```

```
    // Find frequency of each digit temp = num;
```

```
    // Re-initialize freq array to 0 for(i = 0; i < 10; i++)
```

```
    freq[i] = 0;
```

```
    while (temp > 0) { rem =
```

```
        temp % 10;
```

```
        freq[rem]++; temp /=
```

```
        10;
```

```
    }
```

```
    printf("Number: %lld\n", num); for(i = 0; i < 10;
```

```
    i++) {
```

```
        printf("freq_%d = %d", i, freq[i]); if (i < 9) {
```

```
            printf(", ");
```

```
        }
```

```
    }
```

```
    printf("\n"); break;
```

```
default:
```

```
    printf("\nInvalid choice. Please choose 1 or 2.\n"); break;
```

```
}
```

```
return 0;
```

```
}
```